

Investigating flow mechanisms in a forest soil by mixed-effects modelling

C. BOGNER^{a,b}, D. GAUL^c, A. KOLB^a, I. SCHMIEDINGER^a & B. HUWE^a

^aSoil Physics Group, BayCEER, University of Bayreuth, 95440 Bayreuth, Germany, ^bEcological Modelling, BayCEER, University of Bayreuth, Dr.-Hans-Frisch-Straße 1-3, 95448 Bayreuth, Germany, and ^cPlant Ecology, Albrecht-von-Haller Institute of Plant Sciences, University of Göttingen, Untere Karspüle 2, 37073 Göttingen, Germany

Summary

Uniform and preferential flow produces typical infiltration patterns. We made three tracer experiments in a Norway spruce forest soil and qualitatively identified the dominant flow regime based on stained patterns. We analysed soil texture, fine root density and soil bulk density from preferential flow paths and the soil matrix by means of linear mixed-effects models. These models can account for dependences in the data structure caused by hierarchical sampling and can deal with missing values. There were between 44% (topsoil) and 76% (subsoil) larger root densities in preferential flow paths than in the soil matrix. No significant differences in soil texture were detected. The bulk density was greater in the soil matrix by 0.12 g cm^{-3} , which is probably because of a greater soil organic matter content of preferential flow paths. Using flow patterns and model results we identified the dominant flow mechanisms. At this study site, roots constituted the main preferential flow paths and induced macropore flow, especially in the topsoil. In the subsoil, root density decreased and inhomogeneous infiltration from preferential flow paths into the soil matrix caused unstable flow.

Introduction

Water flow in soils can be classified into two categories: uniform flow (Green & Ampt, 1911) and non-uniform or preferential flow (Lawes *et al.*, 1882). The latter group comprises all phenomena where water flows through localized pathways, bypassing a portion of the soil matrix (Hendrickx & Flury, 2001). Non-uniform flow may result in rapid water movement and solute transport to greater depths than predicted by the classical concept of flow through unsaturated soils (as defined by the Richards' equation).

Preferential flow can further be classified into macropore flow, unstable flow and funnel flow (Hendrickx & Flury, 2001). Macropore flow describes water movement along root channels, earthworm burrows, soil fissures, cracks or large inter-aggregate voids in cultivated soils. The impact on groundwater quality may be considerable if preferential flow through continuous macropores becomes a dominant mechanism of transport of agrochemicals to greater depths (Jarvis, 2007). Unstable flow is common in coarse-textured soils and can be induced by variations in texture, water repellency, air entrapment or continuous non-ponding infiltration. Lateral redirection of water caused by textural boundaries is referred to as funnel flow (Hendrickx & Flury, 2001).

Different flow regimes lead to distinct flow patterns. Thus, the flow regime can be identified, at least qualitatively, by using the observed flow patterns that can be visualized by tracer experiments (Flury *et al.*, 1994; Forrer *et al.*, 2000). Weiler & Flühler (2004), for instance, showed typical dye patterns resulting from different flow regimes and classified them on the basis of the width of the stained objects. They used this classification to infer flow types from dye patterns. Kulli *et al.* (2003) performed a hierarchical clustering of dye patterns to distinguish between zones of homogeneous infiltration and zones of preferential flow.

In our study we used the organic food dye Brilliant Blue FCF to trace water flow and define preferential flow paths as stained areas. The above classification of preferential flow suggests that soil texture, root density, water repellency and soil bulk density might be important factors for flow processes in sandy forest soils. Accordingly, the goals of our research were (i) to qualitatively identify the dominant flow regime based on dye patterns and (ii) to analyse whether soil texture, root density and soil bulk density differed significantly between regions of preferential flow and soil matrix. As water repellency might be changed by application of the organic food dye Brilliant Blue, this property was not considered. For the statistical analysis we used linear mixed-effects models, because our data were sampled hierarchically and therefore grouped. Compared with classical linear models, mixed-effects models have the advantage of easily accounting for

Correspondence: C. Bogner. E-mail: christina.bogner@uni-bayreuth.de

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dependences induced in data by hierarchical sampling and can handle heteroscedasticity (or unequal variances) and missing data.

Materials and methods

Site description

The experimental site is located in Waldstein in the Fichtelgebirge (50°08'N, 11°52'E, 770 m above sea level), a mountain ridge in southeast Germany, densely forested by Norway spruce (*Picea abies* L.). The mean annual precipitation is about 1160 mm, and the mean annual air temperature is approximately 5.3°C, with continental winters and warm summers (Foken, 2003). Granites, phyllites and quartzites weathered up to a depth of 30 m constitute the soil parent material (Gerstberger *et al.*, 2004).

The soil is a Haplic Podzol (IUSS Working Group WRB, 2007) with a sandy-to-loamy texture and a mor-like organic layer of up to 15-cm thickness consisting of litter (Oi), fermented (Oe) and humified (Oa) horizons. Five mineral horizons with fluctuating thicknesses within the site can be identified: EA, Bsh, Bs, Bw and Bw/C. Schulze *et al.* (2009) measured extremely acidic pH values (in CaCl₂) ranging between 3.3 in organic horizons and 4.2 in the subsoil. These authors give an overview of mean soil chemical properties. The base saturations varied between 52% in the EA horizon and 40% in the Bsh horizon, which indicates former application of lime (Hentschel *et al.*, 2007). A large deposition of SO₄²⁻ over several decades resulted in a strong acidification of the soil (Gerstberger *et al.*, 2004). Therefore, adjacent forest stands were limed to counterbalance the acidity and the study site accidentally also received an unknown dose of dolomite (CaMgCO₃) discarded from a flying helicopter (Hentschel *et al.*, 2007). At this study site, stones play an important role. The content of rock fragments fluctuates strongly within the site as well as with soil depth and ranges between a few per cent in the topsoil to up to 75 vol% in the subsoil (Gerstberger *et al.*, 2004). The site has been studied for several years within the framework of the Bayreuth Centre of Ecology and Environmental Research (BayCEER) and the main results have been summarized by Matzner (2004).

We worked on three different experimental plots situated approximately 50 m apart at the study site. Especially the EA and the Bsh horizons were often discontinuous and had irregularly shaped boundaries (Table 1). The main difference between the plots was the content of large stones. On plots one and two the content did not exceed 5%. On plot three, however, a continuous layer of stones was present in the Bw horizon with 10–50% and it was partly present in the EA and Bsh horizons.

Tracer experiments

We conducted three tracer experiments with Brilliant Blue FCF and potassium iodide in September and October 2006. Brilliant Blue is a popular tracer in vadose zone hydrology because its colour is readily seen against most soil colours, and it is toxicologically acceptable for environmental use (Flury & Flühler, 1994; Mon *et al.*, 2006). Iodide was used as reference tracer because Brilliant Blue might be strongly sorbed by soil particles

Table 1 Thickness of soil horizons

	Plot 1	Plot 2	Plot 3
Horizon	Thickness / cm ^a		
Organic ^b	13.5 (1.8)	11.3 (0.9)	13.7 (1.4)
AE ^{b,c}	8.3 (2.7)	9.4 (2.9)	6.1 (2.8)
Bsh ^c	4.6 (1.4)	5.2 (1.7)	4.5 (1.9)
Bs	8.6 (1.5)	9.4 (3.0)	9.0 (3.3)
Bw	18.6 (4.1)	19.8 (4.0)	21.7 (5.9) ^d
Bw/C	>44.5	>44.6	>44.1

^aMean values and standard deviations (in parenthesis) of 11 profiles per plot.

^bIrregular, sometimes inclined boundaries.

^cVarying thickness, often discontinuous.

^dBetween 10 and 50% (volume basis) of large stones (>200 mm).

and thus considerably retarded with regard to water flow (Ketelsen & Meyer-Windel, 1999; German-Heins & Flury, 2000; Kasteel *et al.*, 2002).

Tensiometer measurements before the irrigation on plot one indicated a matric potential of −254 hPa at 0.2 m, −216 hPa at 0.4 m and −157 hPa at 0.9 m. On plot two we measured −75 hPa at 0.2 m, −37 hPa at 0.4 m and −20 hPa at 0.9 m and on plot three −35 hPa at 0.2 m, −30 hPa at 0.4 m and −9 hPa at 0.9 m. The irrigation dates were spaced by approximately 1 week. The tracer solution was applied on a surface of about 2 m² with a sprinkler system similar to that described by Ghodrati *et al.* (1990). We did not remove the litter, but took away spruce cones as they covered a large portion of the soil surface. The irrigation rate was 64 mm hour^{−1} and the concentration of the tracers 5 g l^{−1}. Our study is part of a project dealing with the impacts of extreme meteorological events on soil processes, thus the large irrigation rate. For comparison, the maximum 10-minute intensity measured at the study site between 1999 and 2006 was 22 mm and the maximum 1-hour intensity was 54 mm (T. Foken, unpublished data): the irrigation rate was therefore large but not unrealistic. The day after irrigation, 11 vertical soil profiles of 1 m × 1 m were prepared on each plot and photographed with a digital CCD camera in RAW-format. We used a frame of 1 m² and two reference grey scales for geometric correction and white balance adjustment of the photographs.

Sampling and laboratory analyses

Two profiles per plot were treated with an indicator solution of iron(III) nitrate and starch (Lu & Wu, 2003) to visualize iodide infiltration patterns. The nine remaining profiles were sampled alternately for soil texture (three profiles), bulk density (three profiles) and root density (three profiles). We took bulk samples from Brilliant Blue stained and non-stained parts per soil horizon for analysis of soil texture. As some thin horizons were largely bypassed by the flow and could not be sampled, the number of samples differed from plot to plot, resulting in 58 samples altogether. Samples for root density were taken with a soil corer (diameter = 34 mm and height = 50 mm). We could not sample every

soil horizon because some of them were thinner than the diameter of the corer. Therefore we sampled at the defined depth at 0–10 (organic horizons), 10–30, 30–50 and 50–70 cm depth (taking the soil surface as zero). We took three samples from stained areas and three from non-stained ones per profile. At some depths, there was no staining so that the total number of samples was 210. A mean value of root density was calculated per depth and sampled profile, resulting in 71 values. Ninety-six samples for soil bulk density were taken with a soil corer (diameter = 23 mm and height = 10 mm) in stained and non-stained areas at different depths.

Soil samples for analysis of root density were stored for 14 days at 4°C. After washing them in a sieve (mesh size = 0.63 mm), fine roots (diameter < 2 mm, length > 10 mm) were dried at 70°C for 48 hours and weighed. Soil texture was analysed by laser diffraction (Agrawal *et al.*, 1991) using a Mastersizer S (Malvern Instruments Ltd, Malvern, UK) after removal of organic matter by hydrogen peroxide (H₂O₂). To determine soil bulk density, soil samples were dried at 105°C for 24 hours and weighed. Additionally, the soil texture of soil bulk density samples was analysed as described above.

Image analysis

Photographs of soil profiles were taken in the 'lossless' RAW-format so that no information was lost (unprocessed data of the CCD camera sensor), adjusted for white balance and converted to JPEG in Photoshop CS2 (Adobe Systems Inc., San Jose, CA, USA). Then the images were corrected in HALCON (MVTec Software GmbH, Munich, Germany) for perspective and radial distortion. Perspective transformation of the images was obtained via a homogenous transformation matrix (a matrix that combines information about both rotation and translation in one matrix) and radial distortion was modelled by:

$$\begin{pmatrix} u \\ v \end{pmatrix} = \frac{1}{1 + \kappa (\tilde{u}^2 + \tilde{v}^2)} \begin{pmatrix} \tilde{u} \\ \tilde{v} \end{pmatrix}, \quad (1)$$

where κ is the magnitude of the radial distortion, $(\tilde{u}, \tilde{v})^T$ are coordinates of a point in the original image and $(u, v)^T$ are coordinates in the corrected version. If κ is negative, the distortion is barrel-shaped, while for positive κ it is pincushion shaped (MVTec Software GmbH, 2007b).

To classify the images in Brilliant Blue stained and non-stained areas, we transformed them from RGB to HSI (hue saturation intensity) colour space where colour segmentations are robust under varying illuminations (MVTec Software GmbH, 2007a). Mis-classifications caused by the shadow of the metallic frame or large stones, for instance, were corrected manually. The classified binary images (stained parts were coded black and non-stained areas white) were used to calculate the dye coverage (number of stained pixels per depth) in MATLAB (The MathWorks, Natick, MA, USA). We identified the dominant flow regime qualitatively by using the binary images.

Data analysis

Mixed-effects models. Our data were sampled not randomly but hierarchically and are thus grouped on three scales: plot, profile and the depth/horizon. A grouped structure might induce dependences between data because observations that share the same level of grouping might be correlated and different groups might have unequal variances (heteroscedasticity). Classical ANOVA models assume homogeneity of variance, independence of observations and normally distributed residuals and do not allow for missing values. While blocking in ANOVA can handle grouped data, homogeneity of variance (homoscedasticity) can be an issue. Mixed-effects models can easily account for the covariance structure induced by grouping by associating random effects to observations that share the same level of grouping (Pinheiro & Bates, 2000). Homogeneity of variance is not required, but variances can be modelled from data. Predictors can be continuous, factors or both, thus ANOVA, ANCOVA and linear models can be fitted within the same framework. Additionally, mixed-effects models are robust against missing values (Pinheiro & Bates, 2000), which is important for our study. Indeed, as some horizons were largely bypassed by the flow and could not be sampled we have to deal with missing values. A detailed introduction to mixed-effects modelling for ecological data is given in Zuur *et al.* (2009); the mathematical background as well as the implementation in R (R Development Core Team, 2008) are treated by Pinheiro & Bates (2000).

According to the formulation proposed by Laird & Ware (1982) we can write a linear mixed-effects model for a single level of grouping as:

$$\begin{aligned} \mathbf{y}_i &= \mathbf{X}_i \boldsymbol{\beta} + \mathbf{Z}_i \mathbf{b}_i + \boldsymbol{\varepsilon}_i, \\ \mathbf{b}_i &\sim N(0, \boldsymbol{\Psi}), \boldsymbol{\varepsilon}_i \sim N(0, \mathbf{R}_i), i = 1, \dots, M, \end{aligned} \quad (2)$$

where $\mathbf{y}_i$ is the n_i -dimensional response vector for the i th group, $\boldsymbol{\beta}$ is the p -dimensional vector of fixed effects, $\mathbf{b}_i$ is the q -dimensional vector of random effects, $\mathbf{X}_i$ and $\mathbf{Z}_i$ are the known fixed-effects and random-effects regressor (or design) matrices of size $n_i \times p$ and $n_i \times q$, respectively, $\boldsymbol{\varepsilon}_i$ is the n_i -dimensional vector of within-group errors, $\boldsymbol{\Psi}$ the variance-covariance matrix of the random effects, $\mathbf{R}_i$ the variance-covariance matrix of the within-group errors and M the number of groups. The within-group errors are allowed to be heteroscedastic and correlated. Modelling the correlation of the within-group errors allows us to handle time-series and spatial data within the framework of mixed-effects modelling.

In a mixed-effects model, the fixed effects represent parameters of the entire population or certain repeatable levels of experimental factors (such as horizon or defined depth) and are comparable to parameters of a classical linear model. Random effects are associated with individual experimental units drawn at random from a population (such as plots or profiles) and correspond to additional variance terms. The single-level mixed-effects model can be extended to multiple, nested or crossed levels of random effects (Pinheiro & Bates, 2000).

Model building strategy. Mixed-effects models were built in R (R Development Core Team, 2008) using the 'nlme' package (Pinheiro *et al.*, 2008) that estimates model parameters by the restricted (or residual) likelihood (REML). To find adequate models for our data we first built a full mixed-effects model incorporating all fixed effects and random effects for the plot, the profile within plot and the depth or horizon within plot. Then, the significance of random effects was tested against a nested restricted model in a likelihood ratio test (LRT) that is part of the nlme package (Pinheiro *et al.*, 2008). A nested restricted model has a smaller number of random effects but the same fixed effects. The null hypothesis tested in a LRT is that the restricted model fits the data as well as the full model (Pinheiro & Bates, 2000).

Before making inferences about the fixed effects we examined whether the assumptions underlying our models were met: (i) within-group errors that are allowed to be heteroscedastic and correlated are independent for different groups and independent of the random effects and (ii) the random effects are normally distributed, with mean 0 and covariance matrix Ψ (not depending on the group) and are independent for different groups. The within-group residuals provide good surrogates of the within-group errors and can be used in diagnostics plots to check the first assumption qualitatively (Pinheiro & Bates, 2000). We used QQ-plots (quantile-quantile plots) of standardized residuals (raw residuals divided by the estimated within-group error standard deviation). A QQ-plot is a graphical method to compare data with a theoretical distribution (Wilk & Gnanadesikan, 1968). To verify the normality, residual quantiles (sample quantiles) are plotted against quantiles of the normal distribution (theoretical quantiles). We also used scatter plots of standardized residuals versus fitted values or other covariates by group to check whether the residuals are centred at 0 and have equal variances across groups and are independent of the group levels. In our models we used only simple random intercepts. Therefore QQ-plots of random effects were sufficient to assess the second assumption.

At the end of model building we checked for collinearities of the predictor variables. As stated by Shieh & Fouladi (2003), collinearity in a single-level mixed-effects model may inflate the estimates of variance and covariance components. We used the diagnostics proposed by Belsley *et al.* (1980). It calculates variance decomposition proportions from a singular-value decomposition of the design matrix of the fixed effects. A good explanation of the implementation of the method can be found in Belsley (1991).

Variable transformation. The residuals in the mixed-effects model of the root density data were non-normal and we could not model their variance appropriately. Thus, root density data were transformed using the Box-Cox transformation (Box & Cox, 1964) as proposed by Gurka *et al.* (2006) to achieve (an approximate) normality and stabilize the variance:

$$y^{(\lambda)} = \begin{cases} (y^\lambda - 1)/\lambda & \lambda \neq 0 \\ \log(y) & \lambda = 0 \end{cases}, \quad (3)$$

where y represents the root density data and λ the transformation constant. This transformation requires the data to be positive. Hence, we added a constant c to the data because their range included 0. We tried 0.01, 0.1 and 1 as constants and the best fit was achieved with 0.1. We determined λ in a grid search in which we calculated the mixed-effects model for a range of possible values for λ . Then we plotted residual log-likelihood against these values and chose the value that corresponded to the maximum of the curve (Gurka *et al.*, 2006). Once λ was determined, we optimized the constant c in a grid search.

To assess the sensitivity of the estimated fixed effects to the transformation constant λ we followed the advice of Gurka *et al.* (2006) and Lipsitz *et al.* (2000) and compared the inference about the fixed effects between the optimal model with $\hat{\lambda}$ that was found by the grid search and λ equal to 0 (the log-transformation). Using the log-model has the advantage that we can retransform our results to the original scale easily. The mean of a log-normal variable is the exponential of the log mean plus one half of the log variance (Johnson *et al.*, 1994; Manning, 1998; Mullahy, 1998). Because we were concerned about interpretability, we preferred using a convenient value for λ that might introduce a slight bias in the estimated fixed effects (Gurka *et al.*, 2006) to more exact estimates of fixed effects that are difficult to understand.

Goodness-of-fit and simultaneous inference. The parameters of the mixed-effects models are estimated by REML. Thus, the measure of goodness-of-fit R^2 used for least squares estimates is not defined. To assess the overall goodness-of-fit of a mixed-effects model we can use the correlation between the fitted and the observed values (J. Pinheiro, personal communication). Furthermore, we can calculate the percentage of variance explained by fixed effects by comparing the full model with an 'empty' model in which the intercept is the only fixed-effect (Edwards *et al.*, 2008):

$$R^2_{\beta} = \frac{(q-1)v^{-1}F(\hat{\beta}, \hat{\Psi})}{1 + (q-1)v^{-1}F(\hat{\beta}, \hat{\Psi})} \cdot 100\%, \quad (4)$$

where q is the number of estimated parameters in the full model, v is the residual degrees of freedom ($v = n - \text{rank}(\mathbf{X})$, where n is the number of observations and $\mathbf{X}$ the regressor matrix of the fixed effects), and $F(\hat{\beta}, \hat{\Psi})$ is the F statistic for a Wald test of the null hypothesis $H_0: \beta_1 = \beta_2 = \dots = \beta_{q-1} = 0$ so that all parameters except the intercept are zero. This definition allows calculation of the explained variance using a single model fit and thus identical structure of random effects.

After fitting the models, we compared different depths/horizons where it was appropriate. Hothorn *et al.* (2008) extended the theory of simultaneous inference to mixed-effects models. The null hypotheses are formulated through a linear combination of model parameters and are tested simultaneously. Different adjustments of P -values are possible and we used a multiple testing procedure under free combinations (Westfall, 1999). The calculations were made with the add-in package 'multcomp' (Hothorn *et al.*, 2008) in R (R Development Core Team, 2008).

Results

Flow regime

Following the definition by Flüher *et al.* (1996) the expression regime is used to qualify the phenomenology of the flow (preferential or uniform) that can be dominated by one or more physical mechanisms. The patterns with iodide (not shown here) did not indicate any serious retardation of Brilliant Blue with regard to water. Therefore we can associate stained patterns with actual water flow. While the first few centimetres of the litter were homogeneously stained, the flow rapidly condensed into distinct flow paths still in the organic soil horizons (Figure 1). In the upper 20 cm of the profile the flow was dominantly preferential and a large portion of the matrix was bypassed. This region was densely rooted and we observed many blue-stained roots. Between 20 and 60-cm depth, the dye patterns differed considerably among the three experimental plots. On plot one there were distinct spots of dye throughout the profile while on plot two large areas were stained. On plot three, we found a dense stone layer at about 40-cm

depth that formed an obstacle for the flow. Stones reduce the cross-section available for water flow and induce funnelling along their walls. As indicated by the grey shaded area in Figure 1, the amount of staining varied little on this plot and had its maximum in the stone layer. The greatest variation of staining was observed on plot two. On all plots, the dye reached the excavated depth of 100 cm. Despite the differences in patterns, the main common feature of the flow regime in the subsoil is the dispersion of the flow from the preferential flow paths into the matrix.

Modelling results

Content of sand, silt and clay. The composition of the soil fine fraction varied between horizons (Figure 2). Globally the content of sand decreased, the content of silt increased and the amount of clay fluctuated, suggesting a fixed effect for soil horizons. Additionally, the texture fluctuated between the same horizons on different plots, indicating a horizon within plot random effect:

$$y_{ijk} = \beta_j + \beta_{PF} \times PF + b_{ij} + \varepsilon_{ijk},$$

$$i = 1, 2, 3, j = 1, \dots, 5, k = 1, \dots, n_{ij},$$

$$b_{ij} \sim N(0, \sigma_1^2), \varepsilon_{ijk} \sim N(0, \sigma^2), \quad (5)$$

where y_{ijk} is sand or silt in per cent on plot i in horizon j , β_j are the fixed effects for every horizon (1 = EA, 2 = Bsh, 3 = Bs, 4 = Bw and 5 = Bw/C), β_{PF} is the fixed effect for preferential flow, PF is a dummy variable taking the value of 1 for samples from preferential flow paths and 0 for samples from soil matrix, b_{ij} is the horizon within plot random effect, ε_{ijk} is the error term and n_{ij} is the number of observations on plot i in horizon j .

Figure 3(a) indicates a good agreement between fitted and observed values. As an example, we show diagnostic plots for the sand model. The same procedure was applied to all other models to ensure that the model assumptions were met and that points with large residuals did not influence the inference. There is no evidence of heteroscedasticity (Figure 4a) or departure from the normality assumption (Figure 4b,c). Graphs of standardized residuals plotted against horizon or flow region by plot (not shown here), did not suggest any violation, either. There was one observation with an unusually large residual in the Bs horizon, but excluding it from the model did not change the inference about the fixed effects.

The approximate 95% confidence intervals for the random effects and the within-group errors are displayed in Table 2. The correlation between the observed and the fitted values was 0.78 (Figure 3a) and 0.81 (Figure 3b) and the variance explained by fixed effects equalled 39 and 41% for the sand and the silt model, respectively. There were no significant differences between the preferential flow paths and the soil matrix, either in the content of sand ($P = 0.13$) or of silt ($P = 0.76$) (Table 3).

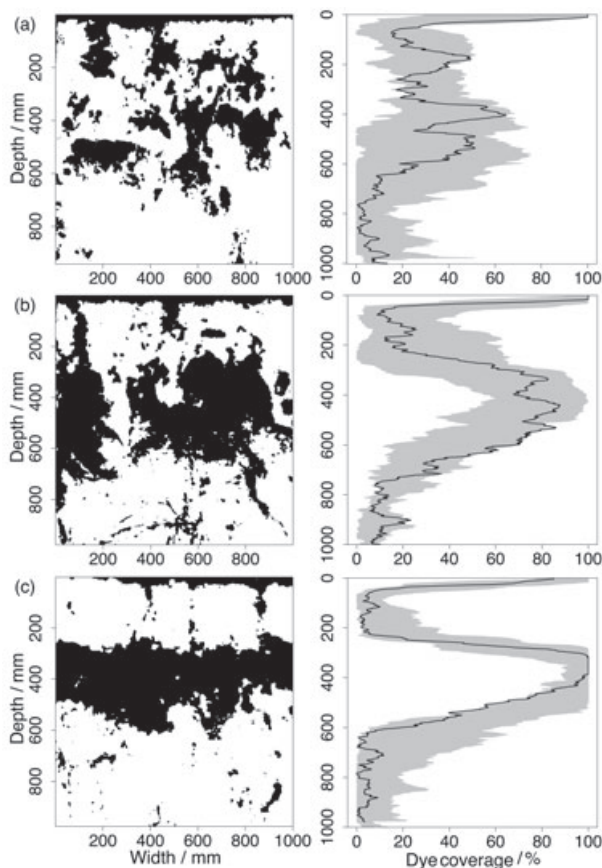


Figure 1 Example images of stained patterns (left) and dye coverage functions (right). From (a) to (c): experimental plot one to three. The grey shaded area indicates the variation between the minimum and the maximum staining calculated from all 11 profiles of a plot and the black line represents the dye coverage of the picture on the left. The depth zero corresponds to the soil surface.

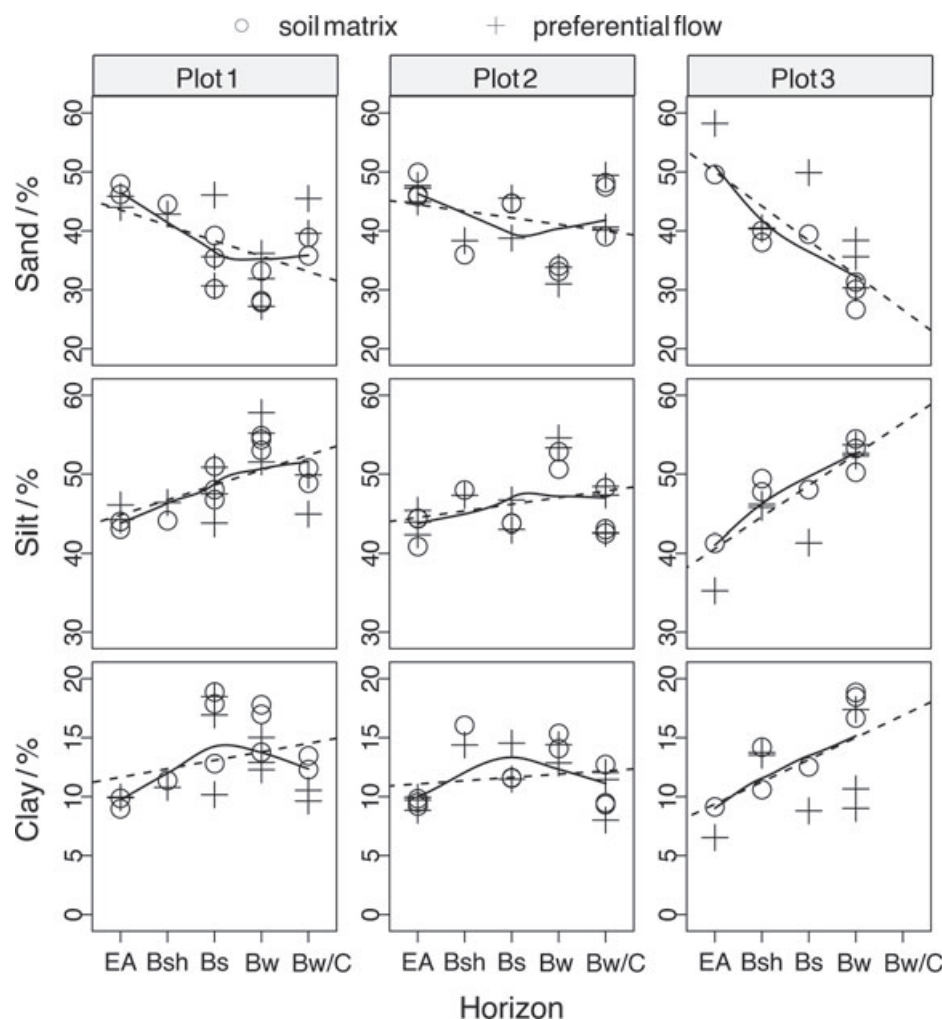


Figure 2 Particle-size distribution of the soil fine fraction with depth (from left to right: plot one to three). Sand fraction is defined as 2000 to 63 μm , silt 63 to 2 μm and clay < 2 μm . Note different scaling of the y-axis. A smooth curve computed by a local regression smoother (loess) (solid) and a linear regression line (dotted) were added to enhance interpretability.

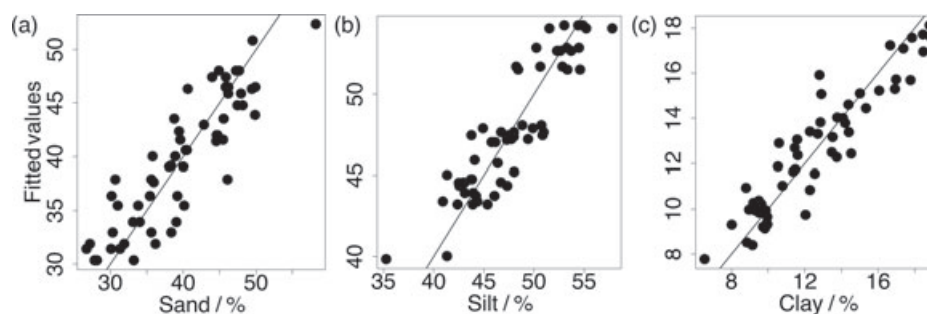


Figure 3 Fitted versus observed values of the (a) sand, (b) silt and (c) clay models (a 1:1 line was added to enhance interpretability).

The content of clay was modelled as:

$$y_{ijkl} = \beta_k + \beta_{PF} \cdot PF + b_{ij} + b_{ik} + \varepsilon_{ijkl},$$

$$i = 1, 2, 3, j = 1, \dots, 4, k = 1, \dots, 5, l = 1, \dots, n_{ijk},$$

$$b_{ij} \sim N(0, \sigma_1^2), b_{ik} \sim N(0, \sigma_2^2), \varepsilon_{ijkl} \sim N(0, \sigma^2), \quad (6)$$

where y_{ijkl} is the clay content as a percentage on plot i in profile j in horizon k , β_k are the fixed effects for every horizon, β_{PF} and PF are the same as in Equation (5), b_{ij} is the profile within plot random effect, b_{ik} is the horizon within plot random effect, ε_{ijkl} is the error term and n_{ijk} the number of observations on plot i in

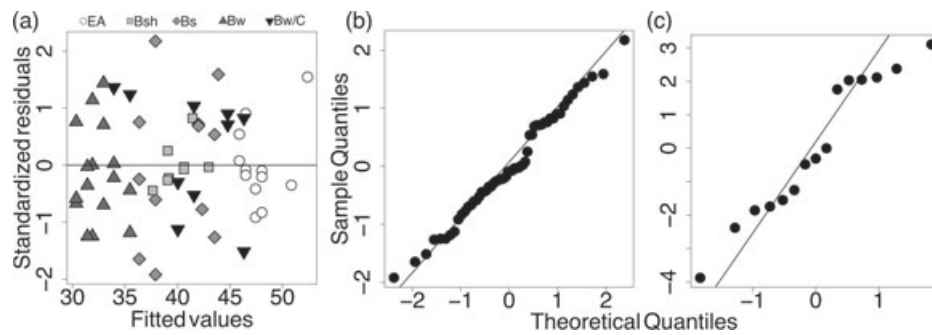


Figure 4 Diagnostic plots for the sand mixed-effects model: (a) standardized residuals versus fitted values per horizon; (b) QQ-plot of the standardized residuals; (c) QQ-plot of the estimates of the horizon within plot random-effect.

Table 2 Estimated random effects and within-group errors and their approximate 95% confidence intervals

Modelled quantity	Horizon random effect			Profile random effect			Within-group error		
	LC ^a	Estimate	UC ^b	LC	Estimate	UC	LC	Estimate	UC
Sand / %	1.6	3.1	6.1	—	—	—	3.0	3.8	4.7
Silt / %	1.0	1.9	3.7	—	—	—	1.8	2.3	2.8
Clay / %	1.1	2.0	3.4	0.4	0.9	2.0	1.1	1.4	1.8
Root density / log (g l ⁻¹)	0.1	0.3	0.5	—	—	—	0.3	0.4	0.5

^aLower confidence limit.

^bUpper confidence limit.

Table 3 Estimated fixed effects and their standard errors for the sand, silt and clay models

Effect	Sand / %		Silt / %		Clay / %	
	Estimate	Standard error	Estimate	Standard error	Estimate	Standard error
EA	47.74***	2.20	42.46***	1.35	9.04***	1.28
Bsh	39.38***	2.32	46.85***	1.42	13.61***	1.31
Bs	40.23***	2.20	45.88***	1.35	13.85***	1.28
Bw	32.00***	2.07	52.95***	1.28	15.59***	1.23
Bw/C	42.41***	2.63	46.32***	1.62	10.86***	1.53
β_{PF}^a	1.54	0.99	-0.19	0.59	-0.62	0.39

*Significant at $P < 0.05$.

**Significant at $P < 0.01$.

***Significant at $P < 0.001$.

^a β_{PF} = Fixed effect for preferential flow.

profile j in horizon k . We excluded three influential points from the fit that increased the correlation between the observed and the fitted values (Figure 3c) from 0.63 to 0.86 without changing the inference about the fixed effects. The percentage of variance explained by fixed effects equalled 28%. The content of clay did not differ significantly ($P = 0.12$) between preferential flow paths and soil matrix (Table 3).

Figure 2 suggests that the texture of successive horizons should be compared. We wanted to know whether the content of sand decreased from the EA to the Bw horizon and increased from the Bw to the Bw/C horizon, and the contrary for the content of silt.

Table 4 Summary of simultaneous inference for the soil texture

	Null hypothesis	Estimate	Standard error
Sand	EA – Bsh ≤ 0	8.37*	3.13
	Bsh – Bs ≤ 0	-0.85	3.13
	Bs – Bw ≤ 0	8.33*	2.94
	Bw/C – Bw ≤ 0	10.51**	3.27
Silt	EA – Bsh ≥ 0	-4.39*	1.92
	Bsh – Bs ≥ 0	0.97	1.92
	Bs – Bw ≥ 0	-7.07**	1.81
	Bw/C – Bw ≥ 0	-6.63**	2.01
Clay	EA – Bsh = 0	-4.57*	1.77
	Bsh – Bs = 0	-0.24	1.78
	Bs – Bw = 0	-1.74	1.71
	Bw/C – Bw = 0	-4.73*	1.91

*Significant at $P < 0.05$.

**Significant at $P < 0.01$.

***Significant at $P < 0.001$.

As there was no clear trend for the content of clay, we tested only whether successive horizons were equal. The sand content did not decrease significantly and the silt content did not increase between the Bsh and the Bs horizons. The clay contents between the Bsh and the Bs and between the Bs and the Bw horizons were not significantly different (Table 4).

Root density. Root density decreased sharply with depth, and varied little between plots (Figure 5a). The grid search for the

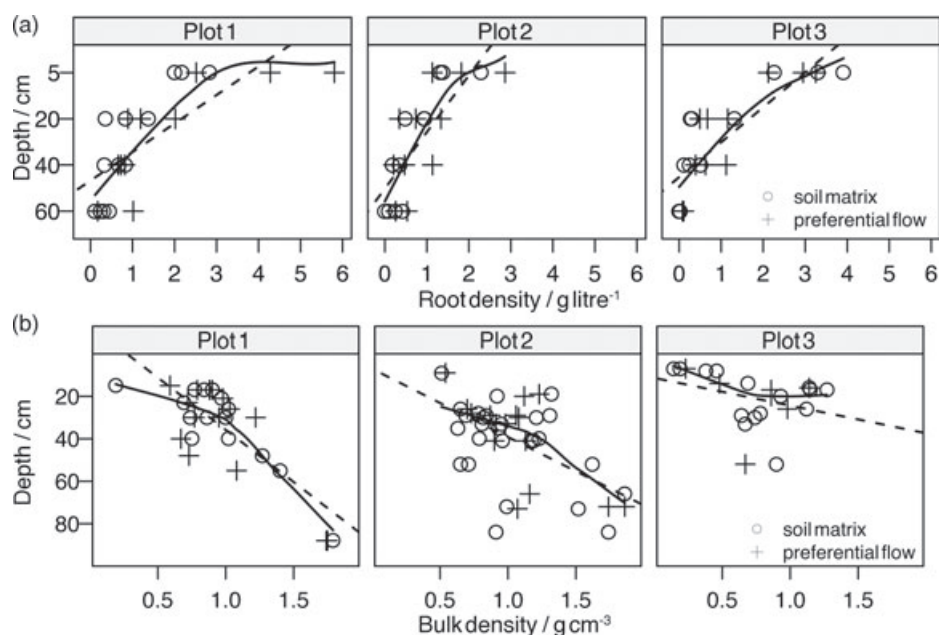


Figure 5 Variation of the root density (a) and of the bulk density (b) with depth.

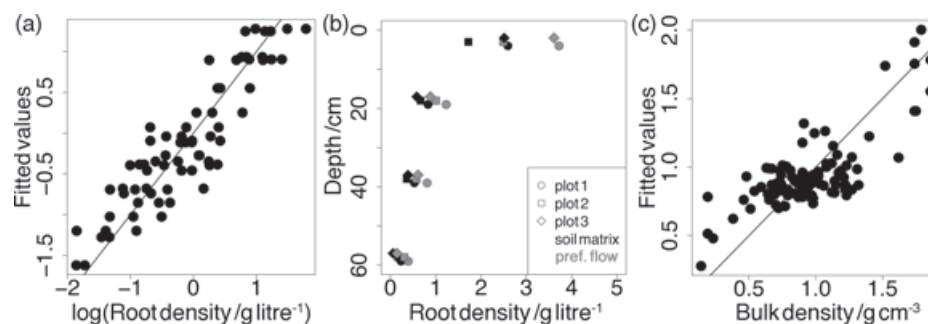


Figure 6 Results of the root density and bulk density models: (a) fitted versus observed root density values (a 1:1 line was added to enhance interpretability); (b) fitted root density values retransformed to the original scale (the depths were slightly jittered for better visibility); (c) fitted versus observed bulk density values (a 1:1 line was added for better interpretability).

Box-Cox transformation resulted in a transformation constant λ of 0.16 and the constant c of 0.156. The comparison between models with λ equal to 0.16 and 0 led to similar estimates of the fixed effects and we kept the convenient model with λ equal to 0:

$$\begin{aligned} \log(y_{ijk} + 0.156) &= \beta_j + \beta_{PF} \cdot PF + b_{ij} + \varepsilon_{ijk}, \\ i &= 1, 2, 3, j = 1, \dots, 4, k = 1, \dots, n_{ijk}, \\ b_{ij} &\sim N(0, \sigma_1^2), \varepsilon_{ijk} \sim N(0, \sigma^2), \end{aligned} \quad (7)$$

where y_{ijk} is the root abundance in g litre⁻¹ on plot i in depth j , β_j is the fixed effects for every depth, β_{PF} and PF are the same as in Equation (5), b_{ij} is the depth within plot random effect, ε_{ijk} is the error term and n_{ijk} the number of observations on plot i in depth j . Depth is a factor of values 'D5', 'D20', 'D40' and 'D60', indicating the middle of the sampled section in centimetres (taking the soil surface as zero). The correlation between the observed

and the fitted values was 0.83 and the variance explained by the fixed effects equalled 57% (Figure 6a). The fixed effect for preferential flow was significant ($P < 0.001$), indicating that the root density in preferential flow paths is higher than in the soil matrix (Table 5).

As the residuals of the root density model are normally distributed and homoscedastic, the retransformation to the original scale on the plot level is straightforward:

$$y_{ijk} = \exp(\beta_j + \beta_{PF} \cdot PF + \hat{b}_{ij} + 0.5 \cdot \hat{\sigma}^2) - 0.156, \quad (8)$$

where y_{ijk} is the root density on the original scale in g litre⁻¹, $\hat{b}_{ij}$ is the estimate of the depth within plot random effect, $\hat{\sigma}^2$ is the estimated within-group error variance and the other elements are the same as in Equation (7). To obtain re-transformed values at the population scale, we replaced the estimate of the random effect $\hat{b}_{ij}$ by $0.5 \cdot \hat{\sigma}_1^2$, $\hat{\sigma}_1^2$ being the estimate of the random-effect

Table 5 Estimated fixed effects and their standard errors for the root density model

Effect	Estimate	Standard error
D5	0.80***	0.18
D20	-0.25	0.18
D40	-0.63***	0.18
D60	-1.28***	0.18
β_{PF}^a	0.34***	0.09

*Significant at $P < 0.05$.**Significant at $P < 0.01$.***Significant at $P < 0.001$.^a β_{PF} = Fixed effect for preferential flow.**Table 6** Summary of simultaneous inference for the root density

Null hypothesis	Estimate	Standard error
D5 – D20 ≤ 0	1.05***	0.25
D20 – D40 ≤ 0	0.38	0.25
D40 – D60 ≤ 0	0.65*	0.25

*Significant at $P < 0.05$.**Significant at $P < 0.01$.***Significant at $P < 0.001$.

variance. At the population scale root density was 44, 49, 54 and 76% greater in preferential flow paths at 5, 20, 40 and 60-cm depths, respectively. Although the difference at 60-cm depth was large, the absolute values of root density were small (Figure 6b). We wanted to know whether root density decreased with depth and simultaneous inference showed that the root density decreased significantly between 5 and 20 cm ($P < 0.001$) and between 40 and 60 cm ($P = 0.011$) (Table 6).

Bulk density. As expected, bulk densities increased with depth. The greatest variability was found on plot two (Figure 5b). On plot three the soil was hardly stained below the stone layer and we sampled less and mostly in the topsoil where the bulk densities were smaller. We tested several models and the best results were achieved with the following analysis of covariance model without any random effect:

$$y_i = \beta_0 + \beta_1 \times \text{sand content} + \beta_2 \times \text{depth} + \beta_{PF} \times PF + \beta_3 \times \text{depth} \times PF + \varepsilon_i, \varepsilon_i \sim N(0, \sigma^2), \quad (9)$$

where y_i is the bulk density, β s are regression parameters (β_0 is the intercept and β_3 the parameter of the interaction term), β_{PF} and PF are the same as in Equation (5), $\text{depth} \times PF$ is the interaction term between the depth and the preferential flow region, and ε_i is the error term. The *sand content* (in %) and the *depth* (in metres) are continuous covariates. We found collinearities between the intercept term, the depth and the sand content. Mean centring the

Table 7 Parameter estimates of the bulk density model

Parameter	Estimate	Standard error
Intercept	1.01***	0.03
Sand content	1.40***	0.34
Depth	1.64***	0.17
β_{PF}^a	-0.12*	0.05
Depth $\times$ PF ^b	-0.81**	0.24

*Significant at $P < 0.05$.**Significant at $P < 0.01$.***Significant at $P < 0.001$.^a β_{PF} = Effect for preferential flow.^bDepth $\times$ PF = Interaction term between the depth and the preferential flow region.

latter two variables (i.e. subtracting the mean) could remove these collinearities.

The fitted values scatter considerably, indicating that there might be some other explanatory variables not included in the model (Figure 6c). As there are no random effects in the model, the only assumption to be verified is that the errors are normally distributed with mean 0 and a constant variance and the diagnostic plots (not shown) did not indicate any violation. The adjusted R^2 of the model equalled 0.58 and the residual standard error 0.22 g cm⁻³. Bulk densities in the preferential flow paths are 0.12 g cm⁻³ smaller than in the soil matrix (Table 7). Additionally, the interaction term between *depth* and *PF* shows that bulk densities in the preferential flow paths increase by 0.81 g cm⁻³ m⁻¹ less with depth than in the soil matrix.

Discussion

Goodness-of-fit

The correlations between fitted and observed values for all mixed-effects models were larger than 0.7 and the percentage of variance explained by fixed effects ranged between 28 and 57%. The random effects were obvious and the hierarchical sampling did actually induce dependencies between data. Thus, the use of mixed-effects models is appropriate. The within-group errors are well estimated and vary at most by a factor of 2 (Table 2). The plots of fitted versus observed values indicate some scatter (for example for root density, Figure 6a), but globally the models represent the data well.

The bulk density model can be improved considerably by incorporating as predictor the content of organic matter removed by hydrogen peroxide (H₂O₂) before texture analysis. This is consistent with previously published results concerning the relationship between soil bulk density and organic matter content or organic carbon content (Manrique & Jones, 1991; Heuscher *et al.*, 2005; Balland *et al.*, 2008). However, the content of organic matter and organic carbon content in preferential flow paths is significantly greater than in the soil matrix (Bundt *et al.*, 2001; Bogner, 2009). Therefore this predictor already incorporates

differences between the flow regions and including it in the model changes the inference about the factor flow region and makes it non-significant.

Flow mechanisms

The dye patterns suggest that at our study site, roots constitute the main preferential flow paths and induce macropore flow, especially in the topsoil. The results of the mixed-effects modelling support this assumption and show that root density is between 44 and 76% greater in the preferential flow paths than in the soil matrix. Using the description of Flühler *et al.* (1996) we can distinguish three flow zones as follows.

- 1 The organic horizons correspond to the 'attractor zone' where water is funnelled towards 'isolated entry ports'. This funnelling is probably caused by the water repellency of the forest floor. Greiffenhagen (2005) reported that infiltration through the forest floor was localized primarily in zones of relatively small water repellency. In contrast, areas with large water repellency were avoided. Experimental results suggest that water repellency is reduced during the winter period, but some water-repellent zones might still persist (Greiffenhagen *et al.*, 2006).
- 2 In the 'transmission zone' (the horizons EA, Bsh and partly Bs) water moves preferentially along root macropores, which can be formed by living as well as decayed roots (Angers & Caron, 1998). However, flow rates through macropores formed by living roots might be reduced in comparison to those shaped by decayed roots (Mitchell *et al.*, 1995). Water probably flows along living roots in a saturated film on the outer root surface or in the space between roots and the pore wall that forms through root shrinkage. After root decay, coatings formed by tissue remnants and the associated microflora still remain on pore walls and enhance water transport (Barley, 1954; Angers & Caron, 1998). The horizons EA, Bsh and Bs have a large content of sand (40–50%) and thus a large conductivity (Rawls *et al.*, 1993). This explains the relatively large stained objects that result from lateral spreading of the dye from preferential flow paths into the soil matrix.
- 3 Comparisons between successive horizons show a significant decrease of root density with depth, which is consistent with the results published by Gaul *et al.* (2008) from the same study site. The authors reported that the first 30 cm of the soil profile contained 80% of the cumulative fine root biomass. The region of small root density coincides with the 'dispersion zone'. Here, the macropores end and water is forced to infiltrate into the soil matrix. As the root system is heterogeneous there is no sharp horizontal boundary between the region where the flow is dominantly preferential and the zone of dispersion into the soil matrix.

The main differences between the three plots were the root density and the stone content. Plot one had the greatest root

density (Figure 5d) and is characterized by isolated stained spots along the whole profile. Here preferential flow is still important in the subsoil horizons and we observed larger roots growing down to the Bw/C horizon. On plot three large stones induce funnel flow along stone walls and infiltration beyond the stone layer is hindered. So the dispersion here is caused mainly by the presence of stones and not the decrease of the root density. Stones are known to increase flow velocities by decreasing the cross-section available for water flow and amplify heterogeneities in soil moisture distribution (Schulin *et al.*, 1987; Hendrickx *et al.*, 1991; Sauer & Logsdon, 2002; Bogner *et al.*, 2008).

Ecological implications

A significant difference in root density and soil bulk density between flow regions does not imply ecological relevance. The greater root densities in the topsoil are ecologically relevant, but the differences in the less rooted lower horizons are probably not. Preferential flow of water along root macropores leads to a better supply of water and nutrients in preferential flow regions than in the soil matrix, especially because of the connection of stemflow and preferential flow. Johnson & Lehmann (2006) described a phenomenon that they called the 'double funnelling of trees'. Rainfall is partitioned by trees into throughfall and stemflow that might be enriched with organic and inorganic nutrients. Stemflow fluxes follow preferentially root-induced macropores below ground, thus bypassing a large portion of the soil matrix. Because of this improved water and nutrient supply, root production might be amplified in preferential flow regions, which is a positive feedback.

The difference in bulk density of 0.12 g cm^{-3} is small and it is difficult to say whether it is important, but it might increase total porosity in preferential flow areas. However, the smaller bulk density probably results from a greater organic matter content in preferential flow paths. It is known that soil bulk density decreases with increasing content of organic matter (Heuscher *et al.*, 2005; Balland *et al.*, 2008). A greater content of organic matter will decrease particle density and might thus counterbalance the effects of a smaller bulk density on total porosity. More organic matter in preferential flow paths probably results from a greater root exudation and decay. Numerous studies have shown that root turnover is an important source of soil carbon and decomposition of dead roots a major input to soil organic matter (Tate *et al.*, 1993; Angers & Caron, 1998; Guo *et al.*, 2005). Additionally, transport of organic matter from topsoil horizons downwards might decrease the bulk density by increasing the soil organic matter content in the preferential flow paths. Kaiser & Guggenberger (2005) analysed the composition of dissolved organic matter during heavy rainstorms and found a good agreement between chemical properties of dissolved organic matter in forest floor leachate and subsoil seepage. They concluded that there might be a direct transfer of organic matter from the forest floor into the subsoil during heavy rainstorms. Thus, smaller bulk densities might partially be attributed to preferential flow.

Conclusions

Mixed-effects models allow the analysis of hierarchically sampled and heteroscedastic data with missing values, where classical modeling approaches such as ANOVA or ANCOVA might be inappropriate. Using dye patterns and mixed-effects modelling we could show that at our study site, roots constituted the main preferential flow paths and induced macropore flow, especially in the topsoil. In the subsoil, root density decreased and inhomogeneous infiltration from ending preferential flow paths into the soil matrix caused non-uniform flow. A large sand content in the topsoil resulted in a strong interaction between preferential flow paths and soil matrix. This explains the relatively large stained objects that resulted from lateral spreading of the dye from preferential flow paths into the soil matrix.

The dye patterns suggest that the organic horizons play a prominent role in initiating preferential flow in the topsoil. The structure of the litter layer as well as water repellency might be important factors. A detailed study of this part of the soil could help to understand which mechanisms force the flow to converge to certain localized flow paths.

Root macropores promote preferential transport of solutes from the organic horizons to the subsoil. Roots are known to strongly influence their immediate environment, the rhizosphere, by exudation of organic compounds. As a consequence distinct chemical compartments might develop with gradients in the transition zone between the soil matrix and preferential flow paths. Moreover, smaller bulk densities in preferential flow paths might result from root exudations and transport of organic matter from the forest floor into the subsoil. Further studies should consider possible impacts of preferential flow on soil chemistry and elucidate whether root production is higher in preferential flow paths.

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